

BatchNorm-Based Weakly Supervised Anomaly Detection in Surveillance Videos Using Optical Flow and RGB Features

A Final Project End Semester Report

submitted by

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BACHELOR OF TECHNOLOGY



**Department of Computer Science and Engineering
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DECLARATION OF ORIGINALITY

I, **Jawahar A S**, with Roll No: **CS21B1031** hereby declare that the material presented in the Project Report titled **BatchNorm-Based Weakly Supervised Anomaly Detection in Surveillance Videos Using Optical Flow and RGB Features** represents original work carried out by me in the **Department of Computer Science and Engineering** at the Indian Institute of Information Technology, Design and Manufacturing, Kancheepuram.

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Jawahar A S

Place: Chennai

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This is to certify that the report titled **BatchNorm-Based Weakly Supervised Anomaly Detection in Surveillance Videos Using Optical Flow and RGB Features**, submitted by **Jawahar A S (CS21B1031)**, to the Indian Institute of Information Technology, Design and Manufacturing Kancheepuram, in partial fulfilment of requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** is a bonafide record of the work done by him/her under my supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

Weakly supervised video anomaly detection (WVAD) aims to identify abnormal events in surveillance footage using only video-level labels, without precise temporal annotations. This makes the task highly challenging, as the model must learn to detect and localize anomalies from coarse supervision. Most existing WVAD methods rely solely on RGB-based features, which effectively capture scene appearance but lack the motion dynamics crucial for identifying anomalous behaviors.

To address this limitation, we propose an enhanced WVAD framework that combines RGB and optical flow features, thereby incorporating both appearance and motion cues. We extract these features using an Inflated 3D (I3D) network, increasing the feature dimensionality from 1024 (RGB-only) to 2048 (RGB + flow), resulting in a richer representation of video segments. To better utilize this high-dimensional input, we introduce a feature enhancement module that leverages an attention mechanism and fully connected layers to highlight informative segments. We further employ a mean-based Pull-Push (MPP) loss, which encourages compact clustering of normal events while pushing anomalous ones apart, improving the model's discriminative power under weak supervision.

The proposed approach was thoroughly evaluated on two widely used benchmark datasets. On the UCF-Crime dataset, it achieved an AUC of 81.65% using both RGB and optical flow features, outperforming the RGB-only baseline, which reached 80.95%. On the XD-Violence dataset, it attained a peak average precision (AP) of 84.60%, marking a clear improvement over the 81.32% AP of the RGB-only variant. These results demonstrate that integrating motion information through optical flow significantly enhances anomaly detection performance in weakly supervised scenarios.

KEYWORDS: Video analysis; Deep learning; Computer vision.

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CHAPTER 1

INTRODUCTION

1.1 Overview

With the widespread installation of surveillance cameras in homes, commercial establishments, and public areas, there has been a growing emphasis on monitoring abnormal activities such as burglary, robbery, and theft. These events are not only rare and unpredictable, but also highly time-sensitive, where early detection can make the difference between prevention and loss. Traditional surveillance systems rely on human operators to continuously monitor live video feeds or review hours of recorded footage, a task that is both mentally exhausting and prone to oversight. In high-risk areas such as banks, retail stores, warehouses, and residential neighborhoods, the ability to quickly identify suspicious behaviors - such as unauthorized entry, lingering near restricted zones, or abrupt movements - becomes critical to initiating timely intervention.

However, these anomalous events are often subtle, brief, and occur in amidst a large amount of routine or uneventful footage, making them difficult to spot without specific cues. This highlights the need for systems that can intelligently and efficiently distinguish between normal and abnormal behavior in surveillance video streams. Without automated support to flag these irregularities in real-time, valuable reaction time may be lost, and crucial evidence might remain buried within massive volumes of video data.

In recent years, deep learning models have revolutionized computer vision by achieving impressive results in object detection, action recognition, and anomaly detection. These models have been successfully applied across various domains, including healthcare, autonomous driving, and industrial monitoring. Similarly, deep learning-based burglary detection models have also been explored, aiming to identify suspicious activities in CCTV footage. However, existing models continue to face challenges in accuracy, frequently producing false alarms or failing to detect subtle anomalies due to the diverse and unpredictable nature of real-world scenarios. Many existing methods

rely on appearance-based features (RGB) alone, which can be insufficient for detecting certain types of anomalies, especially those involving subtle motion patterns. Additionally, weakly supervised methods, which use only video-level labels without precise annotations, introduce further challenges in training robust models.

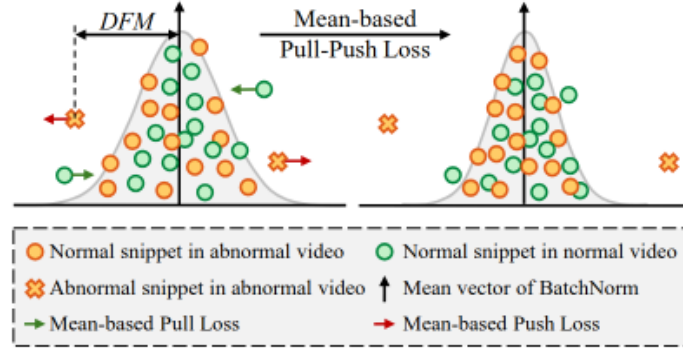


Figure 1.1: Intuition of the proposed DFM criterion and Mean-based Pull-Push (MPP) loss.[1]

To improve anomaly detection performance, our approach builds upon a previous SOTA weakly supervised video anomaly detection (WVAD) framework. The existing method utilizes batch normalization (BN) statistics to structure feature distributions, ensuring that normal features remain clustered while anomalous features are pushed further apart in the feature space. The concept is illustrated in Figure 1.1, where the Mean-based Pull-Push (MPP) Loss leverages the BatchNorm mean vector as a statistical reference. This encourages the separation of anomalous snippets (orange crosses) from normal ones (green circles) by applying a pull force to cluster normal features together and a push force to move abnormal features away. This structured feature representation substantially enhances the model’s capability to differentiate normal events from burglary-related anomalies.

Building on this foundation, we further enhance the input feature representation by incorporating optical flow features extracted from an Inflated 3D (I3D)[3] network. Instead of relying solely on RGB features, optical flow captures motion dynamics, providing additional cues for detecting subtle anomalies that may not be apparent in static frames. By expanding the feature space from 1024 to 2048 dimensions, we aim to improve anomaly detection robustness and generalization across real-world scenarios.

1.2 Objectives

The main objectives of this work are:

- Enhancing weakly supervised video anomaly detection (WVAD) by incorporating optical flow features alongside RGB features to improve anomaly classification.
- Leveraging batch normalization statistics in the loss function to separate normal and anomalous features more effectively.
- Evaluating the impact of optical flow integration by comparing the performance of RGB-only models with multi-modal (RGB + Flow) models on benchmark datasets such as UCF-Crime and XD-Violence.

1.3 Contributions

The key contributions of this work are outlined below:

- We extend an existing WVAD framework by integrating optical flow features alongside RGB features, increasing the feature space from 1024 to 2048 dimensions.
- We retain the BN-based loss function, ensuring that the model effectively learns to separate anomalies from normal events.
- We conduct extensive evaluations to analyze whether integrating motion cues improves anomaly detection accuracy and robustness.

By combining spatial and motion-based features with statistical loss constraints, our approach aims to enhance anomaly detection performance and provide a more reliable framework for real-world surveillance applications.

CHAPTER 2

LITERATURE SURVEY

2.1 Benchmark Datasets for Video Anomaly Detection

Anomaly detection in surveillance videos is a crucial area of research for improving public safety and security. One of the cornerstone datasets in this field is UCF-Crime, introduced by Sultani et al. [4], comprising 1,900 untrimmed real-world surveillance videos that cover 13 categories of anomalies, including robbery, abuse, and explosions. They proposed a multi-instance learning (MIL) approach using only video-level labels, addressing the challenge of weak supervision. Another significant dataset is XD-Violence[5], which expands the scope by including both violent and non-violent anomalies collected from diverse web sources, further pushing the development of robust and generalizable anomaly detection models.

2.2 Existing Approaches in Weakly Supervised Anomaly Detection

Following the introduction of UCF-Crime, several deep learning-based approaches have been developed to improve anomaly detection performance. For instance, RTFM (Robust Temporal Feature Magnitude Learning) [6] focused on robustly learning feature magnitudes to distinguish abnormal snippets from normal ones. This method achieved SOTA performance on UCF-Crime by addressing the limitations of unreliable abnormality criteria. Similarly, UR-DMU (Dual Memory Units with Uncertainty Regulation) [7] introduced dual memory units to explicitly model normality and uncertainty in pseudo labels, demonstrating superior performance on both UCF-Crime and XD-Violence datasets. Another notable contribution is the Self-Supervised Sparse Representation (S3R) [8], which employed self-supervised learning to sparsely represent

video snippets, achieving competitive results by leveraging unlabeled data to enhance feature representation. For scene-aware anomaly detection, Hierarchical Semantic Contrast (HSC)[9] incorporates both foreground object and background scene features with high-level semantics. The method introduces scene-level and object-level contrastive learning to make normal features more compact within the same semantic classes while being separable across different classes. Additionally, Cluster Attention Contrast has emerged as an effective contrastive learning framework that establishes subcategories of normality as clusters. Advancements in the field are marked by the emergence of new sensing technologies, such as UCF-Crime-DVS [10], an event-based dataset that employs dynamic vision sensors to record visual data as discrete events with high temporal precision. Paired with a multi-scale spiking fusion network, this dataset showcases the advantages of event-driven methods in minimizing data redundancy and improving the ability to capture motion dynamics compared to traditional frame-based cameras.

2.3 Optical Flow in Video Analysis

Optical flow estimation has long been essential in video analysis tasks for modeling motion between consecutive frames. One of the earliest techniques, the Lucas–Kanade method[11], estimates motion under the assumption of small and locally constant movement by solving linear equations across a neighborhood. It remains popular for its simplicity and efficiency, but it struggles with large displacements and low-texture regions. Farnebäck optical flow[12] is another classical method that models pixel neighborhoods with polynomial expansion to estimate dense flow. It offers smoother flow fields than Lucas–Kanade but is sensitive to noise and less accurate in complex scenes. To improve robustness and preserve motion boundaries, the Total Variation L1 (TV-L1) approach[13] was introduced. This method formulates flow estimation as an optimization problem with a total variation regularizer, producing sharper flow maps, though it is computationally expensive and often unsuitable for real-time applications.

With the rise of deep learning, methods like FlowNet[14] and RAFT[2] have redefined the landscape of optical flow. FlowNet was one of the first neural architectures trained end-to-end for flow estimation, offering good accuracy at real-time speeds, but

it lacks precision in fine-grained motion. RAFT (Recurrent All-Pairs Field Transforms) further advanced the field by computing dense correlations between all pairs of pixels and refining the flow iteratively with a recurrent unit. It currently achieves state-of-the-art performance on multiple benchmarks. RAFT is available in two versions: RAFT-Large, which prioritizes accuracy, and RAFT-Small, which is faster and more lightweight. In this work, we opt for RAFT-Small to balance computation and performance, making it suitable for real-time anomaly detection in surveillance videos.

2.4 Flow in Anomaly Detection

Optical flow has also been explored as a complementary modality to RGB features for capturing motion dynamics in anomaly detection. For example, the paper "Not Only Look, But Also Listen" [15] integrated optical flow features alongside RGB and audio modalities for violence detection, achieving notable improvements on UCF-Crime and XD-Violence datasets. Optical flow encodes motion patterns, which are crucial for identifying anomalies characterized by unusual movements, such as fighting or running. The study titled "A Classification Method Based on Optical Flow for Violence Detection" [16] proposed a novel feature descriptor called the Histogram of Optical Flow Magnitude and Orientation (HOMO), designed to effectively capture motion-specific anomalies. Such research underscores the critical role of motion features in enhancing the discriminative capability of anomaly detection systems. The HF2-VAD[17] framework innovates through memory-augmented flow reconstruction coupled with flow-guided frame prediction, where anomalies exhibit compounding errors in both stages - poor flow reconstruction degrades subsequent frame prediction, magnifying detection signals. OFST[18] extends this concept with joint optical flow reconstruction and video frame prediction, using divergence between the two pathways as an anomaly metric.

2.5 BatchNorm-based Methods

The BatchNorm-based Weakly Supervised Video Anomaly Detection (BN-WVAD)[1] model represents a recent advancement in this field. Using the statistical properties of

BatchNorm, the authors introduced the Divergence of Feature from Mean (DFM) criterion as a reliable abnormality measure. This approach demonstrated SOTA performance on the UCF-Crime and XD-Violence datasets. The integration of BatchNorm statistics not only improved the reliability of abnormal snippet selection but also addressed the limitations of existing methods, such as sensitivity to label noise and the inability to handle varying abnormality ratios across videos.

2.6 Proposed Fusion Approach

While both optical flow-based methods and BatchNorm-based approaches have shown promise, each has its limitations. Optical flow methods excel at capturing motion dynamics but may struggle with spatial context, while BatchNorm-based methods effectively model normality but may overlook subtle motion-related anomalies. To address these shortcomings, we propose combining optical flow features with RGB features to leverage both spatial and temporal information. By integrating these modalities, we aim to observe better results in anomaly detection, achieving a more comprehensive and robust representation of video snippets. This hybrid approach is expected to enhance the model's ability to detect a wide range of anomalies, particularly those characterized by complex motion patterns or subtle deviations from normal behavior.

CHAPTER 3

METHODOLOGY

Before diving into the complete pipeline of our proposed approach, it's important to first break down the key components that make up the foundation of our method. This chapter takes a closer look at each concept individually, explaining the techniques used for getting optical flow frames, feature extraction, enhancement, and separating normal from abnormal events. By exploring these elements on their own, we can better understand how they contribute to the overall system and work together to improve anomaly detection performance.

3.1 RAFT Optical flow

The Recurrent All-Pairs Field Transforms (RAFT) optical flow algorithm[2] estimates motion by processing two consecutive video frames through a carefully designed pipeline. Initially, both frames are passed through a shared feature extractor network, typically a convolutional neural network, to produce feature maps that are lower in spatial resolution but rich in semantic information. Instead of estimating flow directly from these features, RAFT constructs a high-dimensional correlation volume that encodes the similarity between every pixel in the first frame and every pixel in the second frame. This all-pairs correlation ensures that the model can capture both small and large displacements without relying on traditional pyramidal approaches. Once the correlation volume is built, RAFT initializes a coarse flow field, often set to zero or a basic prior. Then, a recurrent update block - based on a gated recurrent unit (GRU) — iteratively refines this flow field. At each iteration, the update block uses local information from the correlation volume, the current flow estimate, and learned features to predict a residual flow adjustment. By stacking multiple iterations, RAFT allows the flow prediction to become progressively more accurate and detailed.

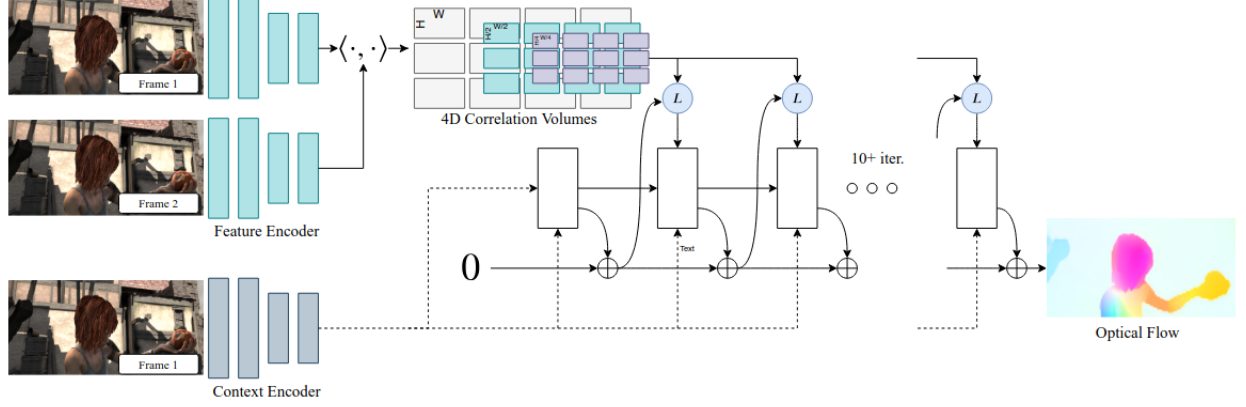


Figure 3.1: Overview of RAFT.[2]

RAFT comes in two variants: the full-sized RAFT model and a lighter version called RAFT-Small. The full model provides state-of-the-art accuracy on standard benchmarks such as KITTI[19] but is computationally demanding, making it less suitable for real-time environments. RAFT-Small addresses this issue by reducing model size and complexity while retaining competitive accuracy. In our work, we employ RAFT-Small to extract optical flow features from surveillance videos, as it offers a practical balance between speed and performance. These flow features are then concatenated with RGB features to enrich the motion representation, which is particularly valuable for detecting anomalies such as sudden movements or physical confrontations in surveillance footage.

3.2 I3D Model

The Inflated 3D (I3D) model[3] is a deep learning-based architecture designed for spatiotemporal feature extraction in videos. It extends standard 2D CNNs into 3D by inflating 2D convolutional filters into 3D kernels, allowing the model to process both spatial and temporal information simultaneously. Unlike 2D CNNs, which analyze frames independently, I3D captures motion patterns across consecutive frames, making it well-suited for video anomaly detection.

For feature extraction, we use two separate I3D models: one trained on RGB frames and another on optical flow sequences. The RGB stream captures appearance-based de-

tails such as objects and background context, while the optical flow stream extracts motion dynamics by estimating pixel displacement between frames. Each model outputs a 1024-dimensional feature vector per video snippet, representing its learned spatiotemporal features.

To take advantage of both appearance and motion cues, we concatenate the RGB and optical flow features, forming a 2048-dimensional representation for each snippet. This richer feature set is then passed through a feature enhancement module to refine temporal dependencies before anomaly classification. The integration of optical flow ensures that the model is more sensitive to motion-related anomalies, improving overall detection performance.

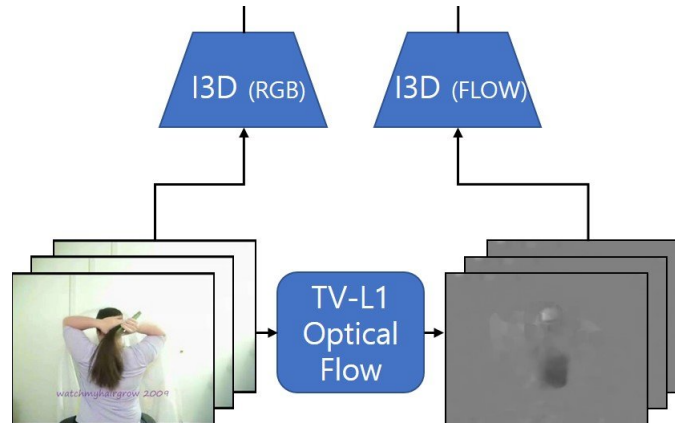


Figure 3.2: The RGB model takes raw frame stacks with minimal preprocessing, while the flow model receives optical flow frames computed using RAFT algorithm.[3]

3.3 Feature enhancer

The feature enhancer plays a crucial role in refining the spatiotemporal features extracted from both RGB and Optical Flow streams before anomaly classification. These features, obtained from a pretrained I3D model, may carry noise, redundancy, or irrelevant temporal variations. To address this, we design the feature enhancer as a lightweight Transformer-based module[20], aimed at capturing meaningful dependencies within short video snippets.

Each feature stream (RGB and Flow) is independently processed by a separate fea-

ture enhancer. The enhancer begins by projecting the input features to a 512-dimensional embedding space via a temporal embedding layer. This is followed by a two-layer Transformer, where each layer includes a multi-head attention block (4 heads, 128 dimensions per head) and a feed-forward network, both preceded by Layer Normalization.

The attention block is designed to capture both content-based and position-aware interactions. In addition to standard dot-product attention, it incorporates a second attention matrix computed using a distance-based positional bias, encouraging localized temporal reasoning. The outputs of both attention paths are concatenated and projected back to 512 dimensions. Each attention output is further refined through a feed-forward network with GELU activation and dropout regularization.

By enhancing the feature quality independently for each modality and before stacking, the model benefits from richer, more discriminative representations that aid downstream anomaly scoring.

3.4 Divergence of Feature from Mean (DFM) Score

The Divergence of Feature from Mean (DFM) Score is a statistical measure that quantifies how much a snippet's feature representation deviates from the batch mean. It acts as an uncertainty measure in weakly supervised anomaly detection, identifying highly divergent snippets that are more likely to be anomalous. DFM is computed using the Mahalanobis distance between a snippet's feature vector and the batch mean obtained from the batch normalization (BN) layer). Since normal snippets tend to be tightly clustered, a higher DFM score suggests greater deviation, making it a useful criterion for selecting samples during training.

The DFM score is given by:

$$DFM(x_i) = \sqrt{(x_i - \mu_B)^T \Sigma_B^{-1} (x_i - \mu_B)} \quad (3.1)$$

where:

- x_i is the feature representation of snippet i .
- μ_B is the mean of the batch normalization layer.
- Σ_B is the covariance matrix of the batch.

3.5 Sample Batch Selection

Since weakly supervised learning lacks precise labels for anomalous snippets, a sample batch selection protocol is used to identify the most informative samples for training. Each training batch consists of an equal number of normal and anomalous video snippets, ensuring balanced learning. The selection process is as follows:

- Video-Level Selection(ρ_s): From each video, the top ρ_s fraction of snippets with the highest DFM scores are selected, as they are likely to be the most discriminative within the video.
- Batch-Level Selection(ρ_b): From the entire batch, the top ρ_b fraction of snippets with the highest DFM scores are further selected, ensuring that globally divergent snippets contribute to training.



Figure 3.3: Illustration of selection strategies in $B=4$ abnormal videos with $T=5$ snippets, using 40% sample-level (ρ_s) and batch-level (ρ_b) selection ratios.[1]

3.6 Mean-based Pull-Push(MPP) Loss

Once the most divergent snippets are selected, they are used to compute the Mean-based Pull-Push (MPP) loss:

- Top-ranked DFM snippets from both normal and anomaly classes are chosen to maintain a balanced contribution.

- Anomalous snippets with high DFM scores are pushed away from the normal cluster, enforcing better feature separation.
- Normal snippets with high DFM scores are pulled closer together, refining the normal feature cluster.

The Mean-Based Pull-Push (MPP) Loss is designed to improve anomaly separation in weakly supervised video anomaly detection by leveraging batch normalization statistics. Instead of using direct anomaly labels, MPP loss structures the feature space by pulling normal features closer together while pushing anomalous features farther away. This is achieved using the batch mean as a statistical reference, ensuring that normal samples remain tightly clustered while anomalies deviate significantly.

The Mean-Based Pull-Push (MPP) Loss is defined as:

$$\mathcal{L}^{\text{mpp}}(X_{\text{dfm}}^n, X_{\text{dfm}}^a, \hat{\mu}, \hat{\sigma}^2) = \frac{1}{K} \sum_{k=1}^K [m + \text{DFM}(X_{\text{dfm}}^n[k]; \hat{\mu}, \hat{\sigma}^2) - \text{DFM}(X_{\text{dfm}}^a[k]; \hat{\mu}, \hat{\sigma}^2)] \quad (3.2)$$

where:

- X_{dfm}^n and X_{dfm}^a are the selected normal and abnormal features with the highest DFM scores.
- $\hat{\mu}$ is the mean vector from the batch normalization layer.
- $\hat{\sigma}^2$ is the variance from the batch normalization layer.
- K is the number of selected normal and abnormal features.
- m is a margin hyperparameter, set to 1 in the implementation.
- $\text{DFM}(X; \hat{\mu}, \hat{\sigma}^2)$ represents the DFM score (Mahalanobis distance) for feature X .

3.7 Model training and optimization

The input features from the feature enhancer are processed through two Conv1D + BatchNorm layers, followed by a Conv1D + Sigmoid layer at the end. Each Conv1D + BN layer extracts hidden features X^{h1} and X^{h2} , contributing to the Mean-Based Pull-Push (MPP) losses $\mathcal{L}_1^{\text{mpp}}$ and $\mathcal{L}_2^{\text{mpp}}$.

The final Conv1D + Sigmoid layer produces the output, where Normality Loss acts as a regularizer:

$$\mathcal{L}^{\text{nor}} = \sum_{b=1}^{B/2} \|C(\text{ReLU}(\text{BN}(X^n[b])))\|_2 \quad (3.3)$$

The overall loss is computed as:

$$\mathcal{L} = \mathcal{L}^{\text{nor}} + \lambda_1 \mathcal{L}_1^{\text{mpp}} + \lambda_2 \mathcal{L}_2^{\text{mpp}} \quad (3.4)$$

3.8 Workflow pipeline

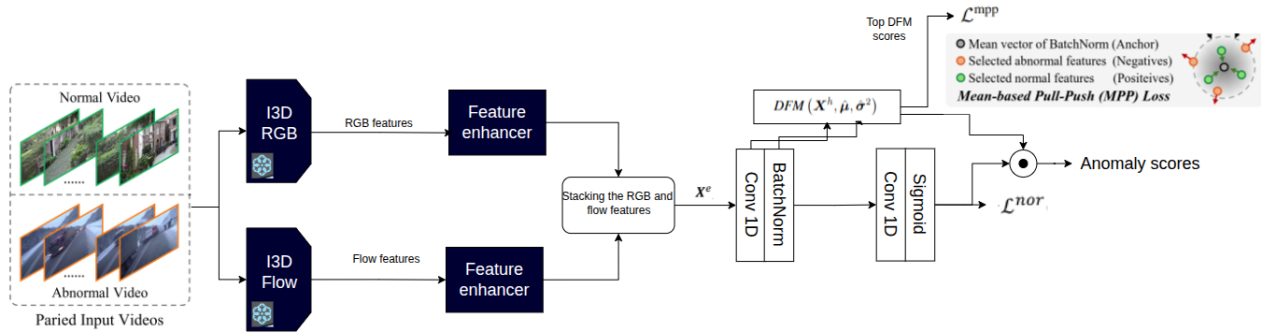


Figure 3.4: The complete workflow of the model involves calculating Divergence of Feature from Mean (DFM) scores during both training and inference.

The training pipeline begins with the creation of mini-batches, each comprising an equal number of normal and anomalous video snippets. Every snippet consists of 16 consecutive video frames. For RGB processing, these snippets are passed through a frozen I3D RGB model, which extracts 1024-dimensional spatial-temporal feature vectors. In parallel, the same input videos are processed using the RAFT-small optical flow algorithm. RAFT computes dense optical flow between consecutive frames, producing high-resolution flow maps. These flow frames are then passed through a frozen I3D Flow model, which generates 1024-dimensional motion-aware features. This dual-stream feature extraction ensures that both appearance and motion cues are captured for each video snippet.

To handle the two feature modalities effectively, we employ two separate Transformer-based Feature Enhancer modules — one for RGB features and one for flow features.

These modules refine the original 1024-dimensional features independently and project them into a lower-dimensional space, each producing a 512-dimensional output. The use of attention in these enhancers helps emphasize critical temporal patterns relevant to anomalous events. After enhancement, the RGB and flow features are concatenated to form a 1024-dimensional unified feature vector that encapsulates both spatial and motion-related cues.

The combined feature vector is then passed through two consecutive 1D convolutional layers with kernel size 1. The first Conv1D layer reduces the dimension to 32, followed by a second Conv1D layer that further reduces it to 16. Both layers are followed by Batch Normalization and ReLU activations to stabilize training and introduce non-linearity. These intermediate representations are critical for computing the DFM (Divergence from Mean) scores and the MPP loss, which play a key role in identifying outlier patterns in the feature space. Finally, a third Conv1D layer with a Sigmoid activation outputs the anomaly prediction score for each snippet. This score is further modulated by the sum of the corresponding DFM values to enhance robustness and accuracy under weak supervision.

CHAPTER 4

EXPERIMENTAL RESULTS

4.1 Hardware details

All experiments were conducted using computational resources provided by the Computer Center at IIITDM Kancheepuram. The system used was equipped with an AMD EPYC 7452 32-Core Processor, comprising 128 logical CPUs (64 physical cores across two sockets) running at 2.35 GHz. For memory, the setup had 32 GB of RAM accessible for our tasks. GPU resources included an NVIDIA A100 (40 GB) dedicated to feature extraction and a Quadro P2000 (5 GB) used for model training. This setup facilitated efficient optical flow computation using the RAFT model and enabled high-throughput training of the anomaly detection pipeline.

4.2 Metrics used

The proposed BN-WVAD framework is evaluated on two prominent WVAD datasets: UCF-Crime and XD-Violence, both of which provide video-level labels. For the UCF-Crime dataset, the primary evaluation metric used is the area under the frame-level receiver operating characteristic (ROC) curve (AUC). This metric measures how well the model differentiates between normal and anomalous frames across different threshold settings, where a higher AUC reflects stronger discriminative capability. In the case of XD-Violence, the frame-level average precision (AP) is used as the key performance measure. AP summarizes the precision-recall trade-off by computing the area under the precision-recall curve, reflecting how well the model retrieves true anomalies without incurring excessive false positives.

4.3 Implementation details

In the experimental setup, an offline feature extraction strategy is employed to enhance efficiency. Videos are initially processed to compute optical flow frames using the RAFT-small model. These flow frames, together with RGB frames, are passed through their respective frozen I3D models to extract 1024-dimensional features. RGB and flow features are stored separately for each video. During training, the model operates on these precomputed features rather than raw video inputs, significantly accelerating training and ensuring consistent input quality.

The model is optimized using the Adam optimizer[21] with a learning rate of 0.0001 and a weight decay of 0.00005. Training is performed for 3000 iterations with a mini-batch size of 64 videos, evenly divided between normal and anomalous samples. The hyperparameters λ_1 and λ_2 are set to 5 and 20, respectively, to balance the different loss components. Selection ratios ρ_s and ρ_b are tailored to the abnormality distribution of each dataset: $\rho_s = 0.1$, $\rho_b = 0.2$ for UCF-Crime, and $\rho_s = 0.2$, $\rho_b = 0.4$ for XD-Violence. During training and inference, multi-crop aggregation is applied to improve prediction robustness, with 10 crops used for UCF-Crime and 5 for XD-Violence, allowing anomaly scores to be averaged over diverse spatiotemporal segments.

To ensure a fair and consistent comparison, the baseline RGB-only model was implemented and evaluated using the same hyperparameters and training settings as described in the reference approach. This reimplementation serves as the benchmark against which the proposed RGB + Flow method is compared. All evaluations, including those on UCF-Crime and XD-Violence, were conducted under identical experimental conditions, ensuring that observed improvements stem solely from the incorporation of flow features.

4.4 Results and comparison

Table 4.1 presents a comparative analysis of the BN-WVAD framework using RGB-only features versus a combination of RGB and optical flow features. On the UCF-Crime dataset, the AUC improves from 80.95% to 81.65% with the addition of flow

features, while on XD-Violence, the AP increases from 81.32% to 84.60%. These consistent gains across both datasets underscore the benefit of incorporating motion cues alongside appearance features.

Table 4.1: Performance comparison of models using RGB and RGB + Flow features

Method	UCF-Crime (AUC %)	XD-Violence (AP %)
RGB Only	80.95	81.32
RGB + Flow	81.65	84.60

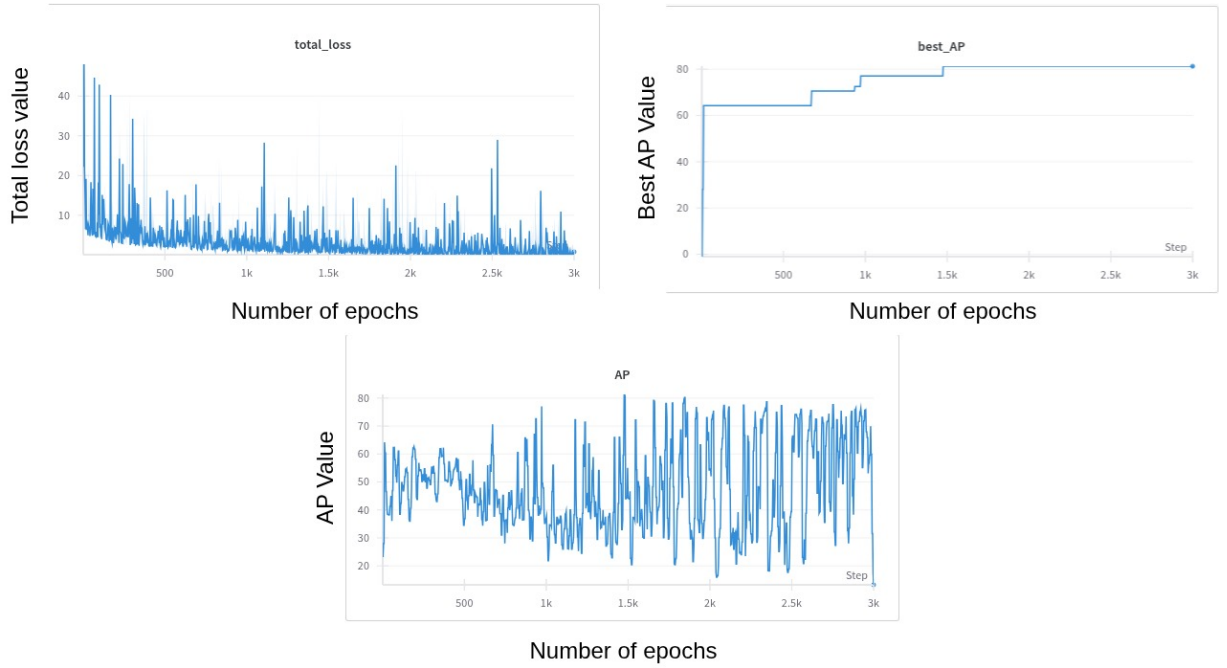


Figure 4.1: Results observed from the XD-violence for using only RGB features. The highest AP achieved is 81.32%.

Figure 4.1 shows the AP trend during training using only RGB features. The curve is highly volatile with large fluctuations and lacks a clear upward trajectory, indicating unstable learning and poor generalization. This suggests that relying solely on appearance features from RGB frames is insufficient for modeling complex temporal dynamics involved in anomaly detection.

In contrast, as seen in Figure 4.2, integrating both RGB and optical flow features leads to a significantly more stable training curve and a much higher peak AP of 84.60%. The optical flow provides crucial motion information that complements the RGB-based spatial features, enabling the model to better distinguish between normal and anomalous events. The overall stability and performance gain underline the impor-

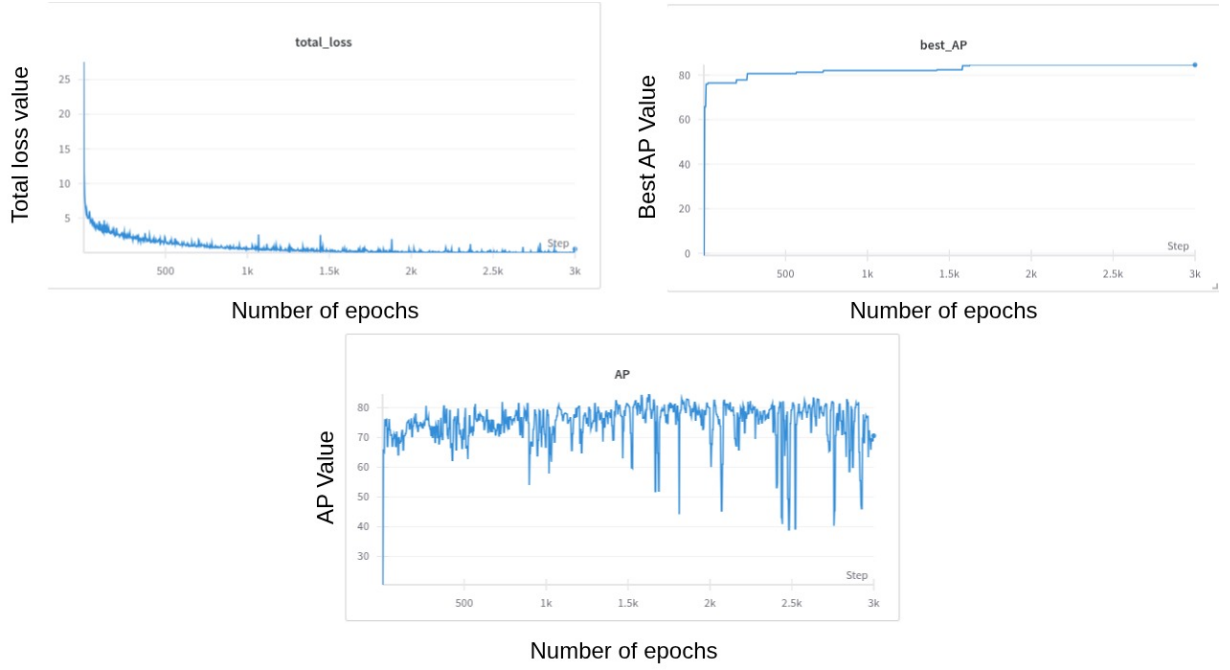


Figure 4.2: Results observed from the XD-violence for using both the RGB and flow features. The highest AP achieved is 84.60%.

tance of incorporating motion cues in temporal anomaly detection.

For the UCF-Crime dataset, a similar trend can be observed but with a smaller performance gap. Using only RGB features, observed from Figure 4.3, the model achieves a maximum Area Under the Curve (AUC) of 80.95%. The AUC plot in this case is relatively smooth, with only minor fluctuations, indicating that RGB features alone offer decent performance with fairly stable training dynamics. The gradual and consistent improvement in the best AUC metric suggests that the model can learn reasonably well from appearance-based features when identifying anomalies.

However, from figure 4.4, when optical flow is added to the input, the model reaches a slightly higher AUC of 81.65%. This gain, while modest, demonstrates that motion information does contribute positively even in datasets where appearance cues are already strong. Interestingly, the AUC curve for the combined RGB and flow model exhibits more noticeable dips, potentially due to noisy or ambiguous motion patterns in non-anomalous segments. Nonetheless, the overall improvement supports the idea that integrating temporal features enhances the model’s capacity to discern subtle anomalies that might be missed using RGB alone.

During training, the total loss and the mean-based Pull-Push (MPP) loss exhib-

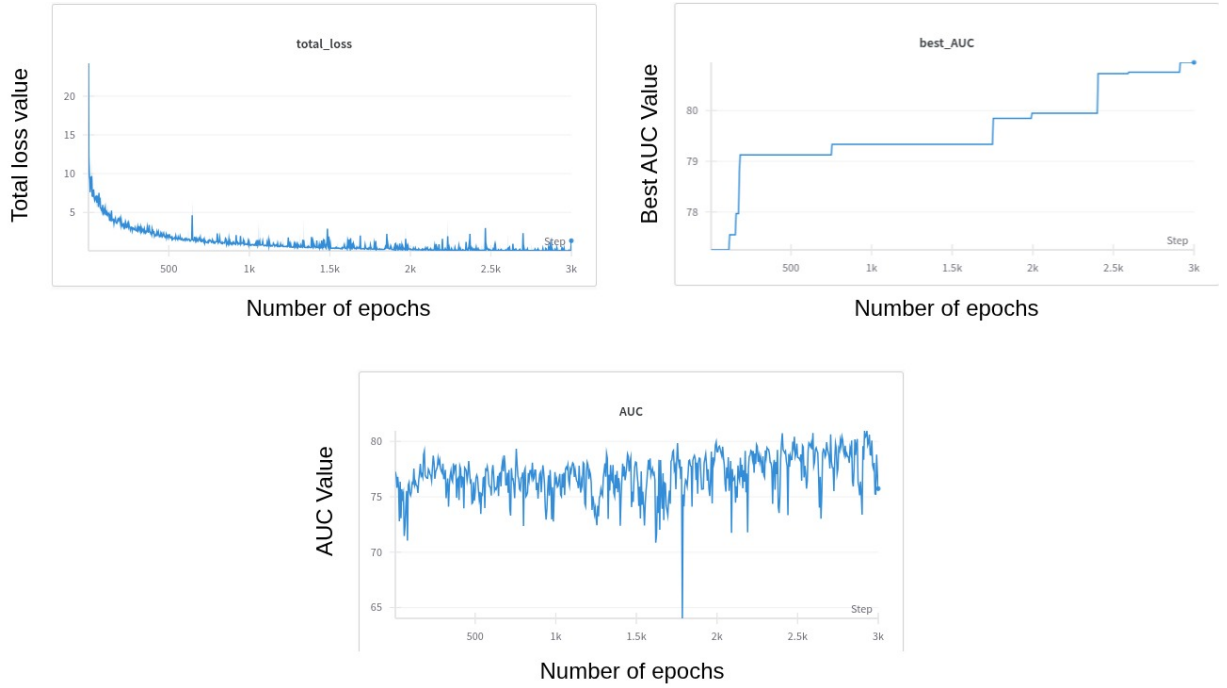


Figure 4.3: Results observed from the UCF-Crime for using only the RGB features. The highest AUC achieved is 80.95%.

ited a steady downward trend, with MPP loss values approaching near zero in several cases. This indicates that the feature space was effectively shaped: features from normal snippets were successfully pulled together, while those from anomalous snippets were pushed apart. Such behavior suggests that the model learned a clean separation in the high-dimensional embedding space. However, despite this encouraging loss profile, the evaluation performance — particularly in terms of AUC and AP — did not reach perfect levels, hinting at potential mismatches between the optimization objective and evaluation criteria.

This discrepancy highlights a fundamental limitation common in weakly supervised video anomaly detection: the loss function, although effective at structuring feature distributions, does not directly optimize for frame-level anomaly localization or ranking, which are crucial for the final metrics. The reliance on video-level labels and snippet-level selection could also lead the model to focus on suboptimal regions during training. Furthermore, the near-zero loss might indicate overfitting to the selected training segments rather than generalizing well to unseen anomalous patterns. This calls for future refinements either in the loss formulation, better snippet selection strategies, or tighter alignment with frame-level objectives to ensure that minimized training loss translates

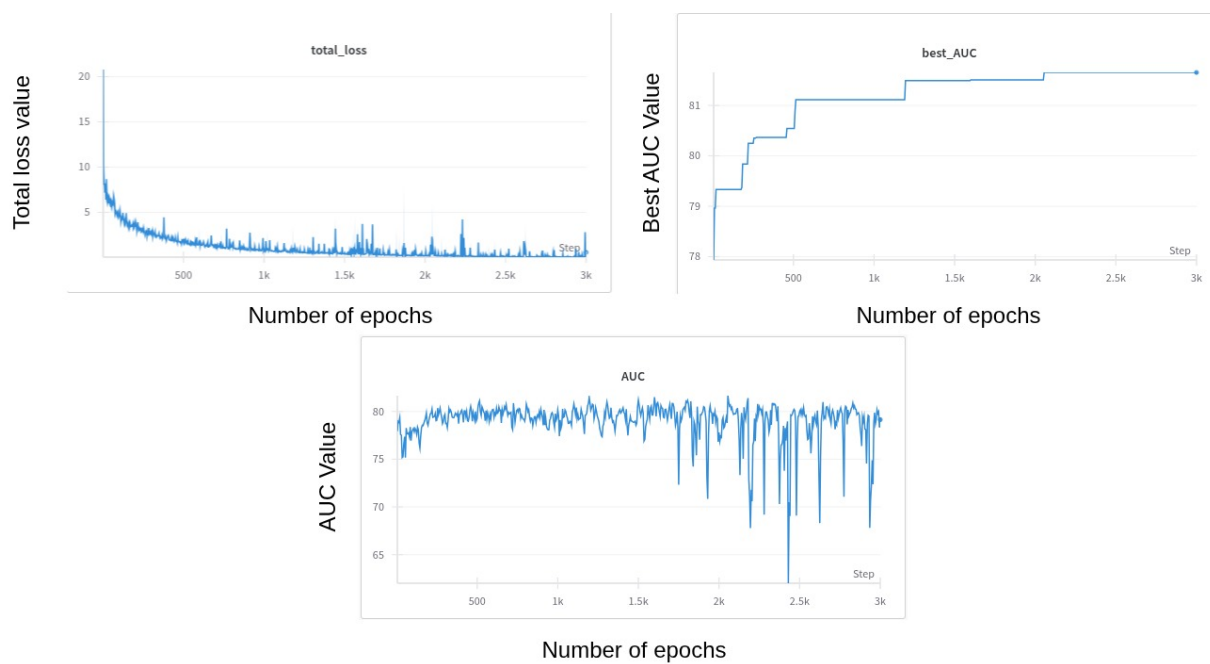


Figure 4.4: Results observed from the UCF-Crime for using both the RGB and flow features. The highest AUC achieved is 81.65%.

more directly into improved real-world performance.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

In this work, we extend a BatchNorm-based weakly supervised video anomaly detection (WVAD) framework by integrating motion-aware optical flow features alongside conventional RGB representations. Instead of relying solely on appearance cues, the proposed approach captures finer motion dynamics by incorporating flow features extracted from video snippets. These features are processed through an additional transformer-based feature enhancer, allowing the system to learn complementary temporal patterns that may not be evident from RGB alone.

By combining RGB and optical flow embeddings into a 2048-dimensional feature vector, the system achieves more expressive and discriminative representations. Experimental results in benchmark datasets demonstrate the effectiveness of this approach, with notable performance improvements over the RGB-only baseline. In the XD-Violence data set, the average precision (AP) increased by more than 3%, while UCF-Crime showed a modest but consistent gain in AUC, highlighting the value of motion information in complex real-world scenes.

5.2 Future Scope

Although the current system demonstrates clear advantages through the incorporation of motion features, there remains potential for further optimization. The use of I3D as a feature extractor introduces significant computational overhead, which limits its suitability for latency-sensitive applications. A promising direction for future work involves replacing I3D with more efficient video encoding architectures such as TimeSformer[22] or MobileNet-based backbones[23], aimed to reduce inference time while preserving

representational quality. Using self-supervised or contrastive pretraining methods could improve model robustness when learning from weakly labeled data, offering better generalization under limited supervision.

A particularly challenging but impactful extension of this work lies in adapting the framework for online or streaming anomaly detection. Transitioning from offline batch processing to real-time sequential analysis would significantly broaden the applicability of the approach, especially in practical deployment scenarios require immediate response.

REFERENCES

- [1] Y. Zhou, Y. Qu, X. Xu, F. Shen, J. Song, and H. T. Shen, “Weakly supervised video anomaly detection via batch normalization-based methods,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 12, pp. 13 642–13 654, 2024.
- [2] Z. Teed and J. Deng, “Raft: Recurrent all-pairs field transforms for optical flow,” in *European Conference on Computer Vision (ECCV)*, ser. Lecture Notes in Computer Science, vol. 12347. Springer, 2020, pp. 402–419.
- [3] J. Carreira and A. Zisserman, “Advancements in action recognition: A new model and the kinetics dataset,” in *CVPR*, 2017, pp. 6299–6308.
- [4] W. Sultani, C. Chen, and M. Shah, “Anomaly detection in real-world surveillance videos,” in *CVPR*, 2018, pp. 6479–6488.
- [5] P. Wu, J. Liu, Y. Shi, Y. Sun, F. Shao, Z. Wu, and Z. Yang, “Not only look, but also listen: Learning multimodal violence detection under weak supervision,” in *European Conference on Computer Vision (ECCV)*. Springer, 2020, pp. 34–51.
- [6] Y. Tian, G. Pang, Y. Chen, R. Singh, J. W. Verjans, and G. Carneiro, “Learning robust temporal feature magnitude for weakly supervised video anomaly detection,” in *ICCV*, 2021, pp. 4955–4966.
- [7] H. Zhou, J. Yu, and W. Yang, “Weakly supervised video anomaly detection via dual memory units and uncertainty regulation,” in *AAAI*, vol. 37, no. 3, 2023, pp. 3769–3777.
- [8] J. C. Wu, H. Y. Hsieh, D. J. Chen, C. S. Fuh, and T. L. Liu, “Self-supervised sparse representation for detecting anomalies in videos,” in *ECCV*, 2022.
- [9] S. Sun and X. Gong, “Hierarchical semantic contrast for scene-aware video anomaly detection,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 22 846–22 856.
- [10] —, “Ucf-crime-dvs: A novel event-based dataset for video anomaly detection with spiking neural networks,” *arXiv preprint arXiv:2503.12905*, 2023. [Online]. Available: <https://arxiv.org/abs/2503.12905>
- [11] B. D. Lucas and T. Kanade, “An iterative image registration technique with an application to stereo vision,” in *Proceedings of the 7th International Joint Conference on Artificial Intelligence (IJCAI)*, 1981, pp. 674–679.
- [12] G. Farnebäck, “Two-frame motion estimation based on polynomial expansion,” in *Image Analysis, SCIA 2003*, ser. Lecture Notes in Computer Science, vol. 2749. Springer, 2003, pp. 363–370.

- [13] C. Zach, T. Pock, and H. Bischof, “A duality based approach for realtime tv-l1 optical flow,” in *Pattern Recognition (Proc. DAGM)*. Springer, 2007, pp. 214–223.
- [14] A. Dosovitskiy, P. Fischer, E. Ilg, P. Hausser, C. Hazirbas, V. Golkov, P. Van Der Smagt, D. Cremers, and T. Brox, “Flownet: Learning optical flow with convolutional networks,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 2758–2766.
- [15] P. Wu, J. Liu, Y. Shi, Y. Sun, F. Shao, Z. Wu, and Z. Yang, “Beyond vision: Utilizing audio-visual learning for multimodal violence detection under weak supervision,” in *ECCV*, 2020.
- [16] J. Mahmoodi and A. Salajeghe, “Violence detection using optical flow and adaptive thresholding techniques,” *Expert Systems with Applications*, vol. 127, pp. 121–127, 2019.
- [17] Z. Liu, Y. Nie, C. Long, Q. Zhang, and G. Li, “A hybrid video anomaly detection framework via memory-augmented flow reconstruction and flow-guided frame prediction,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 13 588–13 597.
- [18] L. Wang, J. Tian, S. Zhou, H. Shi, and G. Hua, “An unsupervised video anomaly detection method via optical flow decomposition and spatio-temporal feature learning,” *Pattern Recognition Letters*, vol. 184, pp. 111–121, 2024.
- [19] A. Geiger, P. Lenz, and R. Urtasun, “Are we ready for autonomous driving? the kitti vision benchmark suite,” in *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012.
- [20] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.
- [21] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” *arXiv preprint arXiv:1412.6980*, 2014.
- [22] G. Bertasius, H. Wang, and L. Torresani, “Is space-time attention all you need for video understanding?” in *Proceedings of the 38th International Conference on Machine Learning (ICML)*. PMLR, 2021, pp. 813–824.
- [23] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, “Mobilenets: Efficient convolutional neural networks for mobile vision applications,” *arXiv preprint arXiv:1704.04861*, 2017.

Weekly Review Report

Roll No: CS21B1031
Name: Jawahar. A.S

Week	Start Date - End Date	Work carried out during the week (with Your signature and Date)	Internal guide's comments with signature and Date
1	6/01/2025 - 12/01/2025	Problem statement Exploration A.S. Loz 10/1/25	
2	13/01/2025 - 19/01/2025	Dataset and Research paper Exploration A.S. Loz 17/1/25	
3	20/01/2025 - 26/01/2025	Research paper exploration related to Anomaly video detection A.S. Loz 24/1/25	weekly
4	27/01/2025 - 02/02/2025	Research paper understanding of Batch-Norm weekly supervised Anomaly Detection A.S. Loz 31/1/25	discussions were held.
5	03/02/2025 - 09/02/2025	Research paper Explanation to guide A.S. Loz 7/2/25	V. m. J.
6	10/02/2025 - 16/02/2025	Understanding the source code and running the code A.S. Loz 14/2/25	
7	17/02/2025 - 23/02/2025	Modifying the code to run on the UCF dataset A.S. Loz 21/02/25	
8	24/02/2025 - 02/03/2025	Optical flow features extraction from the UCF videos (Pt 1) A.S. Loz 1/3/25	
9	03/03/2025 - 09/03/2025	Optical flow features extraction from the UCF (Pt 2) A.S. Loz 3/3/25	
	Mid Semester Review Feedback		

Week	Start Date - End Date	Work carried out during the week (with Your signature and Date)	Internal guide's comments with signature and Date
0	10/03/2025 - 16/03/2025	Optical flow feature extraction from the UCF dataset (pt 3) A.S. Lax 15/3/2025	
	17/03/2025 - 23/03/2025	Optical flow feature extraction from the XD-violence dataset (pt 1) A.S. Lax 22/3/2025	
	24/03/2025 - 30/03/2025	Optical flow feature extraction from the XD-violence dataset (pt 2) A.S. Lax 29/3/2025	
	31/03/2025 - 06/04/2025	Optical flow feature extraction from the XD-violence dataset (pt 3) A.S. Lax 3/3/25	
	07/04/2025 - 13/04/2025	Manipulating dataset and code to incorporate RGB and Flow features A.S. Lax 12/4/25	V. V. A.
	14/04/2025 - 20/04/2025	Changing the model architecture to for better results A.S. Lax 19/4/25	
	21/04/2025 - 27/04/2025	Finetuning hyperparameters to achieve better results A.S. Lax 26/4/25	
	28/04/2025 - 04/05/2025	Visualizing results and exploring model architecture to improve. A.S. Lax 3/4/25	
	05/05/2025 - 11/05/2025	Report Writing A.S. Lax	
	End Semester Review Feedback		

Plagiarism Report

